

Your grade: **100%**

Next item →

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

1. (True/False) In some applications, NMF can make for more human interpretable latent features.

1 / 1 point

- ☒ True
☐ False

✓ **Correct**
Correct! You can find more information in the video Non Negative Matrix Factorization.

2. Which of the following set of features is the least adapted to NMF?

1 / 1 point

- ☐ Word Count of the different words present in a text.
☐ Pixel color values of a an Image.
☐ Spectral decomposition of an audio file.
☒ Monthly returns of a set of stock portfolios.

✓ **Correct**
Correct! You can find more information in the video Non Negative Matrix Factorization.

3. (True/False) The NMF can produce different outputs depending on its initialization.

1 / 1 point

- ☒ True
☐ False

✓ **Correct**
Correct! Please review the video Non Negative Matrix Factorization.

4. Which option is the sparse representation of the matrix below?

1 / 1 point

$[(1, 1, 2), (1, 2, 3), (3, 4, 1), (2, 4, 4), (4, 3, 1)]$

- ☒ $\begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \\ 0 & 4 & 1 \end{bmatrix}$

- ☐ $\begin{bmatrix} 0 & 0 & 1 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \\ 0 & 4 & 1 \end{bmatrix}$

- ☐ $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 2 & 0 \\ 0 & 4 & 2 \end{bmatrix}$

- ☐ $\begin{bmatrix} 0 & 0 & 2 \\ 0 & 3 & 4 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$

✓ **Correct**
Correct! You can find more information in the video Non Negative Matrix Factorization Notebook - Part 1.

5. In *Practice lab: Non-Negative Matrix Factorization*, why did we use "pairwise_distances" from scikit-learn?

- ☐ To calculate the pairwise distance between data points for eliminating outliers.
- ☐ To calculate the pairwise distance between points of the NMF encoded version of the original dataset.
- ☐ To calculate the maximum pairwise distance between points in the dataset.
- ☒ To calculate the pairwise distance between NMF encoded version of the original dataset and the encoded query dataset.

✓ **Correct**

Correct! This helps us determine which existing data point is most similar (and hence the closest) to a new query point.